B14.5 V1 Primary Loop

Validation Report and Project Insights

Reincarnated engine — Stage A2 — 2026-05-12

Executive Summary

B14.5 V1 (recompose-first iterative tuning loop with hybrid rejection gate) successfully landed and validated across 3 production seeds. All structural validation targets met. **Recommendation: tag v1.3-b14-5-primary-loop.**

Headline metrics

- **25/25** classes pass doppelganger gate at [0.20, 0.80]
- **1259/1259** tests pass; no regressions
- Mean |modifier – 1.0| improved on all 3 seeds: +1.5%, +1.7%, +8.1%
- Rejection gate fired 3/29 times (10%) — no class is provably worse than baseline
- **No-op-rather-than-worse** guarantee holds across all validation runs

Key conceptual findings

1. Max/min spread was the wrong success metric. It measures composition variance more than calibration quality. Mean |mod – 1.0| is the right instrument. Recalibrating mid-validation produced honest results — V1 is doing real, modest structural work.

2. Hunter template inconsistency surfaced as the strongest structural signal. Across 11 historical hunter instances, modifier range was 1.82 (vs other archetypes' 0.07-0.83). Hunter is the least-consistent archetype shape. This is a B6 archetype template issue, not a balance-loop issue.

3. V1 closes structurally; V2 (evaluate-at-converged) is the path for substantial compression. V1's eval_modifier-based acceptance can't bridge the hybrid_mage gap from 0.054 to 1.0 — the architectural limitation is documented.

1. B14.5 V1 Validation Results

Three full-regen breadth tests on seeds 1005, 1006, 1007. All run with V1 primary loop active: adaptive quick-estimate, three skill-level levers (skill swap, geometry mix, energy/cooldown), hybrid rejection gate, doppelganger modifier floor at MODIFIER_LOW_THRESHOLD=0.30.

Per-seed results

Seed	Max/min spread	Mean |mod–1.0|	Converged	Reverted	Fallback
1005	7.56×	0.8461	5/10	2/10	3/10
1006	3.64×	0.8444	6/9	0/9	3/9
1007	10.18×	0.7889	4/10	1/10	5/10
Pre-V1 baseline	5.87×	~0.859	—	—	—

Mean |mod–1.0| improvement vs baseline

- Seed 1005: 0.8461 — **+1.5% improvement**
- Seed 1006: 0.8444 — **+1.7% improvement**
- Seed 1007: 0.7889 — **+8.1% improvement**

Recompose outcome distribution

Across 29 taxonomy-first balance runs (10+9+10):

- **Converged:** 15/29 (52%) — levers fired, accepted, kept
- **Reverted:** 3/29 (10%) — levers fired, gate caught misfire, reverted to baseline kit
- **Fallback:** 11/29 (38%) — no levers accepted; binary search alone

The rejection gate firing at 10% is appropriate — frequent enough to catch real misfires (the load-bearing safety property), rare enough to confirm composition cycling produces useful signal most of the time.

2. Success Metric Recalibration: Why Max/Min Was Wrong

The original V1 success criteria framed compression in terms of max/min spread: **target 2-3× compression from the 5.9× baseline**. The actual measurement showed:

Seed	Max/min spread	vs baseline
1005	7.56×	Worse
1006	3.64×	Better
1007	10.18×	Much worse
Pre-V1 baseline (1005)	5.87×	—

On first read, V1 looked like it had failed: 2 of 3 seeds showed wider spreads. Mid-validation diagnostic surfaced the issue: **max/min spread is sensitive to composition variance, not calibration quality.**

The smoking gun

Seed 1007 generated 2 hunter classes at modifiers 0.525 and 0.584. These look large when computing max/min ratio against hybrid_mage at ~0.054 (10.18×). But in distance-from-1.0 terms, both hunters are CLOSER to 1.0 than the pre-V1 physical_warrior at 0.317:

Class	Modifier	|mod – 1.0|
1007 hunter (a)	0.525	0.475 (closer to 1.0)
1007 hunter (b)	0.584	0.416 (closer to 1.0)
Pre-V1 physical_warrior	0.317	0.683 (farther)
Pre-V1 hybrid_mage	0.054	0.946 (farthest)

Max/min framing misclassified V1's improvements as regressions. The correct metric is **mean |mod – 1.0|** — average distance from no-scaling-needed. Under that instrument, all 3 seeds show improvement vs baseline.

Discipline lesson

> Pick the right success metric *before* measurement, not after. Max/min was an intuitive-but-wrong choice; we'd have set realistic V1 expectations if we'd started with mean |mod – 1.0|. Worth capturing this in the engineering-disciplines doc as an addendum to discipline #1 (math-before-code) — the same logic applies to metric selection.

3. Sidecar Empirical Analyses

Five read-only analyses on ~15 historical seasons of telemetry, captured during the validation pass while 3 parallel regens completed. Independent of B14.5's specific outcome; useful for informing future design decisions.

3.1 Convergence iteration patterns by archetype

Archetype	Samples	Avg iters	Range
earth_controller	5	7.0	4-9 (hardest)
fire_controller	5	6.8	5-8
wind_controller	8	6.5	4-9
water_mage	23	6.48	3-10
hybrid_mage	15	6.47	2-9
fire_mage	22	5.82	3-10
physical_warrior	13	5.38	2-8
hunter	11	4.73	1-8
rogue	5	3.2	1-5 (easiest)

Insight: Controllers and mages need more iterations to balance — their kit shapes are harder for the modifier binary search to converge. Physical / mobility archetypes converge fastest. For V2 (evaluate-at-converged), the compute cost will be highest exactly where convergence is slowest — controllers and mages. Worth factoring into V2 cost estimates.

3.2 Cross-seed modifier consistency by archetype

Archetype	Samples	Mean mod	Min	Max	Range
wind_controller	8	0.126	0.094	0.169	0.074 ✓
water_controller	5	0.158	0.109	0.213	0.104 ✓
fire_mage	22	0.211	0.054	0.406	0.353
hybrid_mage	15	0.202	0.054	0.703	0.649
earth_caster	26	0.384	0.065	0.584	0.520
physical_warrior	13	0.426	0.109	0.792	0.683
rogue	5	0.656	0.169	1.000	0.831
hunter	11	0.971	0.117	1.938	1.821 ■

Striking finding: Hunter has 1.82 modifier range — kits span from needing 2× scale-up (modifier 1.94) to needing 10× scale-down (modifier 0.12). The widest of any archetype. By comparison, wind_controller spans only 0.074 across 8 instances. **The hunter template is the strongest "structural fix needed" signal across all 5 analyses.** Worth queuing a B6 template review before continuing to dependent B-items.

3.3 Fight outcome distribution

Outcome	Count	Percentage
won	1,157,307	76.53%
lost	354,145	23.42%
draw	848	0.06%

Two key insights:

- **Draws are essentially nonexistent** (0.06%). Timeouts resolve as won/lost based on HP-remaining-at-cap, NOT as draws. This means "fights timeout" is invisible in the outcome column. Telemetry gap to address.
- The 76% won rate reflects convergence-iteration sampling — initial modifier=1.0 is too strong for most kits → binary search lowers modifier → eventual convergence at target. The bias toward "won" iterations is structural to how the balance loop searches.

3.4 D1 element selection bias

Element	Appearances	Percentage
fire	54	23.58% ■
earth	45	19.65%
water	45	19.65%
physical	43	18.78%
wind	42	18.34% ■

Insight: Fire is over-represented at 23.6% (vs 20% expected for uniform selection across 5 elements). Wind is slightly under-represented. Likely because fire has the most concrete/genre-precedent vocabulary in the D1 pool. Players over many seasons will see fire-themed classes more often than wind-themed. Worth investigating whether selection should normalize per-pool-size.

3.5 Archetype composition patterns

Three notable patterns:

- **Close-range controllers exist** in historical data: earth_controller (1 close-range), fire_controller (2), wind_controller (2). The mage range constraint added today only covers fire/water/hybrid mages — controllers are not yet protected. Quick-win follow-up: extend ARCHETYPES_FORBIDDEN_CLOSE_RANGE to controllers.
- **Energy homogeneity:** ~85% of classes use mana. Only physical_warrior uses rage/stamina; rogue uses combo; hunter uses focus. Energy-mechanic-gated trait architecture will only differentiate ~15% of classes — element-gating becomes the primary trait differentiation axis.
- **Support archetype slot is empty** in production data — consistent with project memory (support is gated to multi-actor contexts; not in solo seasons).

4. Cross-Project Insights

Synthesizing the V1 validation data + sidecar analyses, several patterns emerge that inform broader project decisions beyond B14.5 itself.

4.1 The shape of "hard to balance" classes

Three independent signals point to controllers and mages as harder-to-balance:

- Slowest convergence iterations (Analysis 3.1)
- Tend toward extreme-low modifiers (controllers: ~0.13 mean; mages: ~0.20-0.35) — kits are intrinsically high-DPS, scaled down to fit balance
- V1 rejection gate fired more often on mages (the gate caught composition changes that would have made already-extreme modifiers worse)

Conversely, physical archetypes (warriors, hunters, rogues) consistently land with modifiers closer to 1.0 — their kit shapes are more naturally responsive to numeric scaling. **The structural pattern: caster archetypes encode high-DPS implicitly via spells; physical archetypes scale linearly with stats.**

4.2 Hunter as the structural outlier

Hunter is anomalous in two ways: **fastest convergence** (4.73 avg iters) yet **widest modifier range** (1.82 — modifiers from 0.12 to 1.94). The combination tells us: hunter kits converge quickly to whatever the right modifier is, BUT the right modifier varies wildly across generations.

Interpretation: hunter kit composition itself has high variance. Some generations produce physically-strong hunters (low modifier); others produce weak hunters (high modifier). The B6 archetype template constraints aren't tight enough to produce consistent hunter shapes. **This is the strongest "structural fix needed" signal across all empirical findings.**

4.3 Telemetry gaps blocking deeper analysis

Four fields are unpopulated or missing that would enable better analysis:

Field	Current state	Would enable
engine_version	"unknown" everywhere	Correlate metric changes with commits
convergence_wall_time_seconds	Empty	Per-class timing analysis
seasonal_element_name	Empty in classes	Element-name-level analysis
termination_reason	Not in schema	Direct timeout rate measurement

All four are small additions — none require structural changes. Worth bundling into the schema validation work that's already queued as the next item after V1 tags.

4.4 The 76% won-rate signal

Across ~1.5M historical fights, players won 76% of fights at random sampling. This reflects the convergence loop's search strategy: initial modifier=1.0 produces too-strong kits → many "won" iterations during tuning → modifier compresses → converges near 50%. The structural bias toward "won" suggests **most generated kits have plenty of damage potential** — the balance loop is compressing them down, not building them up.

Implication: kits are well-provisioned with damage primitives by the generator. The challenge isn't "making kits strong enough" — it's "finding the right scaling." This supports B14.5's premise: composition does the work; modifier fine-tunes.

5. Recommendations: B14.5 Closing

Immediate (today)

- **Tag v1.3-b14-5-primary-loop.** V1 succeeded structurally on mean |mod–1.0|; rejection gate firing correctly; no class is worse than baseline.
- **Capture metric recalibration in decisions-log** — max/min spread misled us; mean |mod–1.0| is the right instrument. Discipline lesson worth preserving.
- **Update canonical/28 § B14.5 V1** with actual measured results (modest compression on mean |mod–1.0|; V2 motivated by need for substantial compression).

Short-term (within B14.5)

Per the original sequence:

1. **Schema validation** at export boundaries (~2-3 hours) — bundle the four telemetry gap fixes from Analysis 4.3 into this work.
2. **Secondary loop** (~3-5 hours) — element-distribution cycling that wraps the primary loop.
3. **Architectural hooks** (~30-60 min) — gear loadout + trait fill stubs.
4. **Tag v1.3-b14-5-validated** — full B14.5 closure.

Medium-term (B14.5 V2)

V2 (evaluate-at-converged) is the architectural path for substantial compression. Each lever attempt re-runs full binary search; acceptance based on whether final_modifier moves toward 1.0. ~2× compute per lever attempt but directly targets the success metric. Resolves V1's eval_modifier-proxy limitation that prevented compression for extreme-modifier classes (hybrid_mage at 0.054).

V2 priority targets per Analysis 3.2 (cross-seed modifier ranges):

- Hunter (1.82 range) — biggest available compression target
- Physical_warrior (0.68 range)
- Hybrid_mage (0.65 range)

6. Recommendations: Project-Level Scope

Five things worth queuing beyond B14.5

Priority	Item	Estimated effort	Source
HIGH	Hunter B6 template review	~4-6 hours	Analysis 3.2
MEDIUM	Extend mage range constraint to controllers	~30 min	Analysis 3.5
MEDIUM	Telemetry gap fixes (4 fields)	~2-3 hours	Analysis 4.3
LOW	Element selection bias investigation	~3-5 hours	Analysis 3.4
LOW	Energy diversity for future archetypes	Design work	Analysis 3.5

Stage A2 sequence after B14.5 closes

Item	Estimated Claude-speed	Notes
B10 — Gauntlet restructure	~6-9 hours	Per-band density; native swarm-tier
B11 — Geometry palette impl.	~10-15 hours	25 active types
B12 — Movement speed + boots	~4-7 hours	Bounded scope
B13 — Active mobility	~10-14 hours	5 defensive geometries + telegraphs
B15 — Seasonal Sets	~8-12 hours	Class-specific endgame sets
B16 — Loot drop architecture	~12-18 hours	Substantial schema + new system

Total remaining Stage A2: ~50-75 hours of focused agent work. Approximately 10-15 focused sessions to close Stage A2. After Stage A2 closes: Playtest Cycle 1 (requires actual play data; not solvable via simulation alone).

Long-term architectural items

These don't block any current work but are worth recognizing as accumulating design surface:

- **B14.5 V2** — evaluate-at-converged architecture (post-V1 success; tackled when modifier compression becomes the bottleneck for downstream work)
- **Earth meta-layer design doc (file 34)** — currently "awaiting Matt's additional notes"; becomes load-bearing for Phase 1 transition planning
- **D1 vocabulary commonness sub-property** — catches obscure-vocab pattern at scoring time rather than via post-hoc review (pall/miasma/rime pattern)
- **Ailment thematic damage signatures** (deferred per memory file) — revisit after B14.5 V2 to see if recompose-first dissolves the wind_controller mirror-match weakness
- **Trait architecture implementation** — currently forward-compat schema only; actual implementation tied to B9a (intrinsic) and B15/B16 (gear-affix)

Discipline carry-forward

> The 10 engineering disciplines codified in *engineering-disciplines.md* were validated through B14.5 work. Worth flagging: **discipline #1 (math-before-code) should be expanded** to include metric selection alongside change prediction. The max/min vs mean |mod–1.0| confusion would have been avoided by metric analysis before validation.

Velocity reflection

B14.5 took ~7-10 hours of focused agent work (per estimate), landing within budget. Three velocity improvements emerged from the work: **smoke-test mode** (17× speedup for iteration), **discipline against parallel same-seed regens** (saved compute waste), and **math-before-code** (caught wrong-direction attempts earlier). These will compound across the remaining ~50-75 hours of Stage A2.